Module-8: Computer Vision

Practice Set

1. Environment Setup + Segmentation, Threshold, Filtering, Features & Alignment

1. What are the steps to install OpenCV in Python?
2. What is image segmentation? Give one example.
3. Explain image thresholding. Difference between simple and adaptive threshold.
4. What is image filtering? Why is it needed?
5. What are image features? Give examples.
6. What is image alignment and where is it used?

---

2. Read/Write Image + Edge, Motion, Stitching, Histogram, Contours, FFT

1. How do you read and save an image using OpenCV?
2. What is edge detection? Name two edge detection methods.
3. What is motion detection? Give one real-life use case.
4. Explain image stitching and panoramas.
5. What is a histogram? Why is it useful in images?
6. What is Fourier Transform in image processing?

---

3. Resize & Rotate Image + Object Classification, Tracking & Detection

1. Why do we resize images in computer vision?
2. What is the difference between resize and scale?
3. How does image rotation work?
4. What is object classification?
5. Difference between object detection and object tracking.
6. Give an example where object tracking is required.

---

4. Thresholding, Filtering, Edge, Stitching, HDR, Histogram + PyTorch Vision

1. Explain different types of image thresholding.
2. What is Gaussian filtering and why is it used?
3. What is edge detection used for?

4. What is HDR imaging?
5. How do we plot an image histogram using Matplotlib?
6. What is PyTorch Vision? Name two pre-trained models.

---

5. Color Spaces & Morphological Transformations

1. What is a color space? Explain RGB and HSV.
2. Why do we convert RGB to HSV?
3. What is erosion in image processing?
4. What is dilation? How is it different from erosion?
5. What are contours in an image?
6. How do you find and draw contours?

---

6. Template Matching, Image Pyramids, Addition & Blending

1. What is template matching? Give one use case.
2. Explain image pyramids and their types.
3. Why are image pyramids used?
4. What is image addition?
5. Difference between image addition and blending.
6. Where is image blending used in real life?

---

7. Video Processing, Face Detection, MeanShift & CamShift

1. How do you capture video from a camera using OpenCV?
2. How can you read and play a video file?
3. How do you extract frames from a video?
4. What is face detection? Which algorithm is commonly used?
5. What is MeanShift?
6. Difference between MeanShift and CamShift.

---

8. Object Tracking, Pose Estimation, Features, KNN & Pretrained Models

1. What is object tracking?
2. What is pose estimation?

3. What are feature detectors? Name two.
4. Explain feature matching.
5. How does digit recognition using KNN work?
6. What are pre-trained vision models? Why are they useful?